

COSC 594

Variational Inference (III)

Dongsheng Ding

Department of Electrical Engineering and Computer Science

University of Tennessee, Knoxville

Outline

- KL DIVERGENCE
- EVIDENCE LOWER BOUND
- POLICY OPTIMIZATION

Outline

■ KL DIVERGENCE

Dual representation of KL divergence

$$D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) = \sup_{f(\cdot)} \mathbb{E}_{x \sim q(\cdot)} [f(x)] - \log \mathbb{E}_{x \sim p(\cdot)} [e^{f(x)}]$$

■ UPPER BOUND

$$\begin{aligned} D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) &:= \mathbb{E}_{x \sim q(\cdot)} \left[\log \frac{q(x)}{p(x)} \right] \\ &\leq \sup_{f(\cdot)} \mathbb{E}_{x \sim q(\cdot)} [f(x)] - \log \mathbb{E}_{x \sim p(\cdot)} [e^{f(x)}] \end{aligned}$$

$$f(x) = \log \frac{q(x)}{p(x)} \quad - \quad \text{a specific choice}$$

■ LOWER BOUND

$$\begin{aligned}D_{\text{KL}}(q(\cdot) \parallel p(\cdot)) &= \mathbb{E}_{x \sim q(\cdot)} \left[\log \frac{q(x)}{p(x)} \right] \\&= \mathbb{E}_{x \sim q(\cdot)} \left[\log \frac{q(x) \mathbb{E}_p [e^{f(x)}]}{p(x) e^{f(x)}} \right] + \mathbb{E}_{x \sim q(\cdot)} [f(x)] - \log \mathbb{E}_p [e^{f(x)}] \\&= \mathbb{E}_{x \sim q(\cdot)} \left[\log \frac{q(x)}{p'(x)} \right] + \mathbb{E}_{x \sim q(\cdot)} [f(x)] - \log \mathbb{E}_p [e^{f(x)}] \\&= D_{\text{KL}}(p(\cdot) \parallel p'(\cdot)) + \mathbb{E}_{x \sim q(\cdot)} [f(x)] - \log \mathbb{E}_p [e^{f(x)}] \\&\geq \mathbb{E}_{x \sim q(\cdot)} [f(x)] - \log \mathbb{E}_p [e^{f(x)}]\end{aligned}$$

$$p'(x) = \frac{p(x)e^{f(x)}}{\mathbb{E}_p[e^{f(x)}]} \quad - \quad \text{an exponentially weighted distribution for any } f$$

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ELBO maximization

$$\text{ELBO : } \text{ELBO}(p) := \mathbb{E}_{x \sim q(\cdot)} \mathbb{E}_{z \sim q(\cdot | x)} \left[\log \frac{p(x, z)}{q(z | x)} \right]$$

■ EQUIVALENCE

$$\begin{aligned} & \underset{p(\cdot, \cdot)}{\text{maximize}} && \text{ELBO}(p) \\ & && \Updownarrow \\ & \underset{p(\cdot, \cdot)}{\text{minimize}} && D_{\text{KL}}(q(\cdot, \cdot) \parallel p(\cdot, \cdot)) \\ & && \Updownarrow \\ & \underset{p(\cdot, \cdot)}{\text{minimize}} \quad \underset{f(\cdot, \cdot)}{\text{maximize}} && \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [f(x, z)] - \log \mathbb{E}_{(x, z) \sim p(\cdot, \cdot)} [e^{f(x, z)}] \end{aligned}$$

Convex-concave minimax optimization

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Fenchel conjugate

$$F^*(g) = \sup_{f \in \mathcal{F}} \langle f, g \rangle - F(f)$$

- ★ convexity
- ★ a special case w/ $\mathcal{F} = \{f(x, z)\}$, $g = q$

$$\underbrace{D_{\text{KL}}(q(\cdot, \cdot) \| p(\cdot, \cdot))}_{:= F^*(q)} = \sup_{\substack{f(\cdot, \cdot) \\ \text{red}}} \mathbb{E}_{(x, z) \sim q(\cdot, \cdot)} [\text{red } f(x, z)] - \underbrace{\log \mathbb{E}_{(x, z) \sim p(\cdot, \cdot)} \left[e^{\text{red } f(x, z)} \right]}_{:= F(f)}$$

KL divergence is a **Fenchel conjugate function**

Conjugate of conjugate function is itself

$$(F^*)^* = F$$

$$\Downarrow$$

$$F(f) = (F^*)^*(f) = \sup_{g \in \mathcal{G}} \langle f, g \rangle - F^*(g)$$

■ KL DIVERGENCE REGULARIZATION

$$\underbrace{\log \mathbb{E}_{(x,z) \sim p(\cdot, \cdot)} \left[e^{f(x,z)} \right]}_{:= F(f)} = \sup_{q(\cdot, \cdot)} \mathbb{E}_{(x,z) \sim q(\cdot, \cdot)} [f(x, z)] - \underbrace{D_{\text{KL}}(q(\cdot, \cdot) \| p(\cdot, \cdot))}_{:= F^*(q)}$$

Donsker and Varadhan's variational formula

Fenchel conjugate of KL divergence $\equiv$ KL divergence regularization

Policy optimization

$$\pi^+(\cdot | s) \leftarrow \underset{\pi'}{\operatorname{argmax}} \langle Q^\pi(s, \cdot), \pi'(\cdot | s) \rangle - D_{\text{KL}}(\pi'(\cdot | s) \| \pi(\cdot | s))$$

$$\pi^+(\cdot | s) \propto \pi(\cdot | s) e^{Q^\pi(s, \cdot)}$$

★ a special case w/ $q = \pi'$, $f = Q^\pi$

$$\underbrace{\log \mathbb{E}_{a \sim \pi(\cdot | s)} \left[e^{Q^\pi(s, a)} \right]}_{:= F(Q^\pi)} = \sup_{\pi'(\cdot | s)} \mathbb{E}_{a \sim \pi'(\cdot | s)} [Q^\pi(s, a)] - \underbrace{D_{\text{KL}}(\pi'(\cdot | s) \| \pi(\cdot | s))}_{:= F^*(\pi')}$$

★ KL divergence minimization

$$\underset{\pi'}{\operatorname{minimize}} \quad D_{\text{KL}}(\pi^+(\cdot | s) \| \pi'(\cdot | s))$$

Summary

■ KL DIVERGENCE

■ EVIDENCE LOWER BOUND

ELBO maximization $\equiv$ KL divergence minimization

■ POLICY OPTIMIZATION

policy optimization $\equiv$ KL divergence minimization